

3RD YEAR INDUSTRIAL TRAINING PROJECT

Project: Sales Forecasting & Business Insights



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Project Objective & Problem Statement



Problem Statement

Retailers collect huge sales datasets but lack automated analytics pipelines to track demand and performance.



Core Objective

Build an end-to-end Python sales analytics and forecasting pipeline using real transactional data.



Expected Outcome

Deliver product/regional insights and a Linear Regression machine learning model for Q1 2026 sales.

Dataset Overview & Key Parameters

TOTAL SALES REVENUE

₹12.32 Cr

₹12,32,02,383.00 Total Revenue

VOLUME FULFILLED

3,769 Units

Cumulative quantity sold

AVERAGE PRODUCT PRICE

₹33,346.47

Weighted mean realization

HISTORICAL HORIZON

12 Months

Jan 2025 – Dec 2025

Data Fields & Schema

date: YYYY-MM temporal index field

product: Categorical SKU identifiers

quantity_sold: Unit transaction volume

price: Unit sale price in INR (₹)

region: Distribution zones (4 areas)

sales: Computed metric (quantity × price)

Business Performance Summary

OVERALL REVENUE

₹12,32,02,383

✓ ₹12.32 Crore total sales

VOLUME FULFILLED

3,769 Units

Total products delivered

CATALOG PRICE RANGE

₹33,346.47

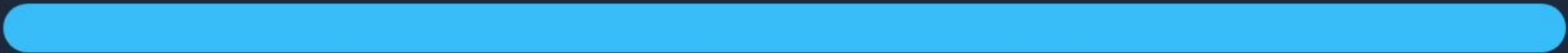
High-ticket electronics avg



Formula & Logic: $\text{Sales} = \text{quantity_sold} \times \text{price}$ — High-ticket consumer electronics drive strong overall revenue across 3 primary categories.

Sales Performance by Product

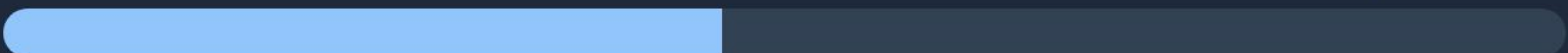
Laptop (Top Product) ₹5,99,45,089 (48.66%)



Tablet ₹3,55,83,711 (28.88%)



Phone ₹2,76,73,583 (22.46%)



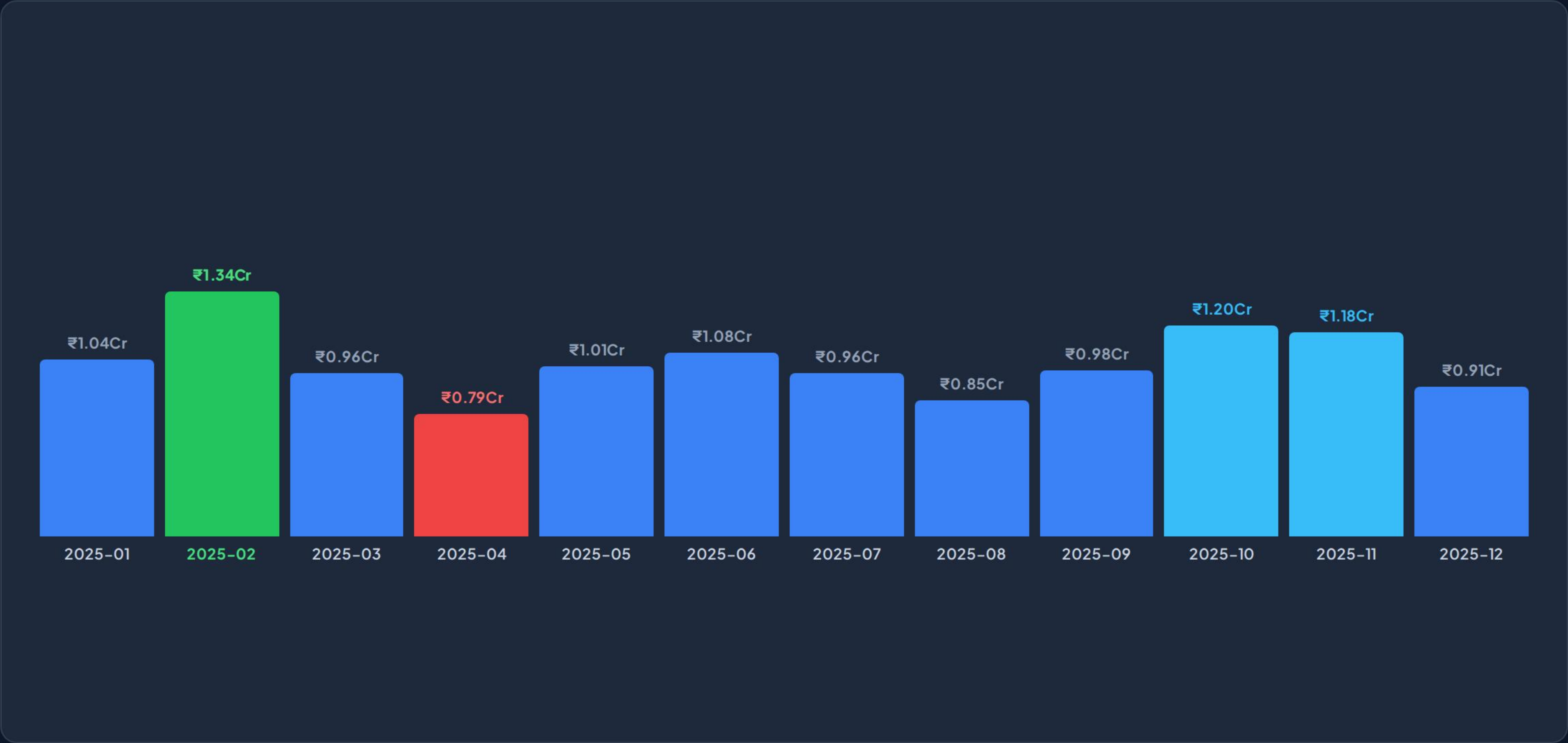
Dominant Category

Laptops generate nearly half (48.66%) of total company revenue (₹5.99 Crore), outperforming Tablets and Phones combined.

Sales Performance by Region



Monthly Sales Breakdown (2025 Historical Data)



Best Performing Highlights & Trend

HIGHEST SALES MONTH

February 2025

₹1,33,66,655 (₹1.34 Cr)

TOP SELLING PRODUCT

Laptop

₹5,99,45,089 (₹5.99 Cr)

TOP SALES REGION

West Region

₹3,31,61,750 (₹3.32 Cr)

OVERALL TREND TRAJECTORY

Decreasing Slope

First vs last month drawdown (Jan ₹1.04Cr → Dec ₹0.91Cr)

Machine Learning Sales Forecasting

Model Setup

Algorithm: Scikit-Learn `LinearRegression()`

Feature (X): Sequential Month Index (1 to 12)

Target (y): Monthly Sales Volume in INR (₹)

3-MONTH FORECAST (Q1 2026)

Month	Projected Revenue (₹)
2026-01	₹1,00,95,520 (₹1.01 Cr)
2026-02	₹1,00,69,160 (₹1.01 Cr)
2026-03	₹1,00,42,800 (₹1.00 Cr)

Actual Sales vs. Forecast Sales Comparison



Business Insights & Recommendations

1. Inventory Management

Prioritize inventory stock for Laptops as they generate over 48% of total company revenue (₹5.99 Cr).

2. Regional Expansion

Boost marketing campaigns in the East region to bridge the revenue gap with top-performing West (₹3.32 Cr).

3. Seasonal Planning

Prepare for peak demand cycles during early Q1 (February) and late Q4 (October–November).

PROJECT CONCLUSION

Conclusion & Project Summary

Pipeline Implementation

End-to-end data processing, aggregation, visualization, and ML forecasting successfully implemented in Python.

Technology Stack

Pandas, NumPy, Matplotlib, Scikit-Learn utilized for data transformation and Linear Regression modeling.

Project Status

Project Completed Successfully.
Ready for presentation and evaluation.